

An Analysis of Support vector machine

Support vector machine

Support vector machine -- Part 1

SVM and the hinge loss Recall that the (soft-margin) SVM classifier $\hat{w}, b: x \mapsto \text{sgn}(\hat{w}^T x - b)$ is chosen to minimize the following expression:

Support vector machine -- Part 2

Requires full labeling of input data Uncalibrated class membership probabilities—SVM stems from Vapnik's theory which avoids estimating probabilities on finite data The SVM is only directly applicable for two-class tasks. Therefore, algorithms that reduce the multi-class task to several binary problems have to be applied; see the multi-class SVM section. Parameters of a solved model are difficult to interpret.

Support vector machine -- Part 3

where x_i is a training sample with target value y_i . The inner product plus intercept $\langle w, x_i \rangle + b$ is the prediction for that sample, and ϵ is a free parameter that serves as a threshold: all predictions have to be within an ϵ range of the true predictions. Slack variables are usually added into the above to allow for errors and to allow approximation in the case the above problem is infeasible.

Support vector machine -- Part 4

$$f^* = \arg \min_{f \in H} \epsilon(f) + R(f).$$

Support vector machine -- Part 5

This is called the dual problem. Since the dual maximization problem is a quadratic function of the c_i subject to linear constraints, it is efficiently solvable by quadratic programming algorithms. Here, the variables c_i are defined such that

Support vector machine -- Part 6

Thus, for large values of C , it will behave similar to the hard-margin SVM, if the input data are linearly classifiable, but will still learn if a classification rule is viable or not.

Support vector machine -- Part 7

The w and b that solve this problem determine the final classifier, $x \mapsto \text{sgn}(w^T x - b)$, where $\text{sgn}(\cdot)$ is the sign function. An important consequence of this geometric description is that the max-margin hyperplane is completely determined by those x_i that lie nearest to it (explained below). These x_i are called support vectors.

Support vector machine -- Part 8

of test examples to be classified. Formally, a transductive support vector machine is defined by the following primal optimization problem: Minimize (in w, b, y)

Support vector machine -- Part 9

$$b = w^T \phi(x_i) - y_i = [\sum_{j=1}^n c_j y_j \phi(x_j) \cdot \phi(x_i)] - y_i = [\sum_{j=1}^n c_j y_j k(x_j, x_i)] - y_i.$$
$$\begin{aligned} b = w^T \varphi(x_i) - y_i &= \left[\sum_{j=1}^n c_j y_j \varphi(x_j) \cdot \varphi(x_i) \right] - y_i \\ &= \left[\sum_{j=1}^n c_j y_j k(x_j, x_i) \right] - y_i. \end{aligned}$$

Support vector machine -- Part 10

minimize w, b, ζ
$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \zeta_i$$
 subject to $y_i (w^T x_i - b) \geq 1 - \zeta_i, \zeta_i \geq 0 \forall i \in \{1, \dots, n\}$
$$\begin{aligned} & \underset{\{w, b, \zeta\}}{\text{minimize}} \\ & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \zeta_i \\ & \text{subject to } y_i (w^T x_i - b) \geq 1 - \zeta_i, \zeta_i \geq 0 \text{ for all } i \in \{1, \dots, n\} \end{aligned}$$

Support vector machine -- Part 11

Note that f is a convex function of w and b . As such, traditional gradient descent (or SGD) methods can be adapted, where instead of taking a step in the direction of the function's gradient, a step is taken in the direction of a vector selected from the function's sub-gradient. This approach has the advantage that, for certain implementations, the number of iterations does not scale with n , the number of data points.

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Support vector machine -- Part 12

$$\mathbf{z} \cdot \mathbf{w} - b = \sum_{i=1}^n c_i y_i k(\mathbf{x}_i, \mathbf{z}) - b \quad \text{where } \mathbf{z} \in \mathcal{H} \text{ and } \mathbf{w} \in \mathcal{H} \text{ is the weight vector.}$$

Support vector machine -- Part 13

$$\epsilon(f) = E[\ell(y_{n+1}, f(X_{n+1}))] \quad \text{where } \ell(y, f(x)) = \max(0, 1 - y f(x)).$$

Support vector machine -- Part 14

$$y_i (\mathbf{w}^T \mathbf{x}_i - b) = 1 \quad \text{where } \mathbf{w}^T \mathbf{x}_i - b = y_i \text{ for } i = 1, \dots, n.$$

Support vector machine -- Part 15

$$\mathcal{D} = \{ \mathbf{x}_i \in \mathcal{R}^p \mid \mathbf{x}_i \in \mathcal{R}^p, i = 1, \dots, k \}$$

Support vector machine -- Part 16

The original maximum-margin hyperplane algorithm proposed by Vapnik in 1963 constructed a linear classifier. However, in 1992, Bernhard Boser, Isabelle Guyon and Vladimir Vapnik suggested a way to create nonlinear classifiers by applying the kernel trick (originally proposed by Aizerman et al.) to maximum-margin hyperplanes. The kernel trick, where dot products are replaced by kernels, is easily derived in the dual representation of the SVM problem. This allows the algorithm to fit the maximum-margin hyperplane in a transformed feature space. The transformation may be nonlinear and the transformed space high-dimensional; although the classifier is a hyperplane in the transformed feature space, it may be nonlinear in the original input space. It is noteworthy that working in a higher-dimensional feature space increases the generalization error of support vector machines, although given enough samples the algorithm still performs well. Some common kernels include:

Support vector machine -- Part 17

where the y_i are either 1 or -1, each indicating the class to which the point $\mathbf{x}_i$ belongs. Each $\mathbf{x}_i$ is a p -dimensional real vector. We want to find the

"maximum-margin hyperplane" that divides the group of points $\mathbf{x}_i$ for which $y_i = 1$ from the group of points for which $y_i = -1$, which is defined so that the distance between the hyperplane and the nearest point $\mathbf{x}_i$ from either group is maximized. Any hyperplane can be written as the set of points $\mathbf{x}$ satisfying